VALIDATION AND AI USE RECORD — IREN LIMITED (NASDAQ: IREN)

Student: Elliot | Course: FIN 43900 | Sourced Ledger, Documented Sensitivity Experiments, and AI Disclosure

1. Decision, Version, and As-of Boundary

Decision: WATCH / DEFER at $12.12/sh intrinsic value vs $45.73 market price. | **Commit:** bf079d8 | **Date:** 2026-10-08 | **Environment:** Python 3.13.15, pandas 3.0.6, numpy 2.5.3, streamlit 1.65.0, reportlab 5.0.1.

2. Data and Convention Ledger

Material Input	Sourced Value	SEC Filing Source Locator	Treatment / Risk Handling
Total Revenue (FY26)	$707,007 thousand	10-K, Cons. Stmt. of Operations, p. F-7	Audited GAAP revenue. Excludes unearned $4.0B ARR.
Cash Cost of Rev (Power)	$219,706 thousand	10-K, Cons. Stmt. of Operations, p. F-7	Reconciles to 68.92% gross margin excl. D&A.;
Audited Cash Capex	$4,333,143 thousand	10-K, Stmt. of Cash Flows, p. F-9	Includes $2,998M PP&E; + $1,335M hardware (missed by aggregators).
Customer Prepayments	$1,842,546 thousand	10-K, Cons. Balance Sheet, p. F-6	Microsoft cash advance; non-dilutive working capital liquidity.
Ending Cash Balance	$5,895,591 thousand	10-K, Cons. Balance Sheet, p. F-6	Direct filing input; verifies capex pre-funding.
Term Debt & Notes	$7,836,740 thousand	10-K, Notes 22 & 23, pp. F-43–F-47	Carrying debt deducted in enterprise-to-equity bridge.
Basic Ordinary Shares	394.059 million	10-K Cover Page (August 14, 2026)	Primary share count denominator reflecting actual current shares.

3. Validation Register & Refusal Gates

Claim / Test	Expected Behavior	Actual Result	Disposition
Synthetic Known-Answer	Lab 09 ABG engine yields exactly $291.75/sh.	Implied VPS = $291.75; Gap = 0.0000 across all 5 yrs.	**PASS** (Reconciled to cent)
Double-Entry Articulation	Assets - Liab - Equity = 0.0000 every year.	Gap = 0.0000 in FY27E-FY31E; cash ties to BS.	**PASS** (Articulated)
Refusal Gate (Cash Break)	Inject +$500M artificial cash break in FY28E.	Script halted: 'MODEL REFUSAL: Gap = $+500.0M'	**PASS** (Refusal triggered)
Stress Negative FCFE	Capitalizing negative terminal FCFE is refused.	Terminal value flagged UNAVAIL; refused Gordon model.	**PASS** (Economic logic held)
README Cold-Run Test	Cold execution reproduces visible output.	py run_analysis.py ran in 1.8s; exported visible_output.json.	**PASS** (Reproduced)

4. Documented Changed-Input Sensitivity Experiments & Directional Verification

Preserved Commits: 3125ff9, 0644c82 | **Documented Hypotheses & Outputs from Lab Workflow**

Tested Driver	Old → New Value	Hypothesized Direction	Preserved Model Output	Outcome & Verification
Cash Gross Margin	70.0% → 75.0% (+5 pp)	EBIT & value must rise monotonically; capex keeps FCFE negative.	FCFF VPS rose from $12.12 to $14.61 (+$2.49/sh). FCFE VPS rose from -$5.31 to -$2.93.	**MATCHED**. Margin expansion helps but is secondary to capex. Action: WATCH-DEFER.
Revenue Growth Path	Base → +5 pp/yr path	Revenue scales sharply; VPS increases substantially.	FCFF VPS rose from $12.12 to $19.25 (+$7.13/sh). FCFE VPS crossed to +$0.99.	**MATCHED**. Confirms Capacity Energization is dominant driver. Action: WATCH-DEFER.
FY31 Single-Year Growth	10.0% → 11.0% (+1 pp)	Minimal impact on overall 5-year present value.	FCFE VPS: -$5.3145 → -$4.9213 (+$0.3933/sh change).	**MATCHED**. Confirms single-year terminal growth is secondary to capex timing.

5. Named AI-Use Register

Tool / Assistant Surface	Date	Task	Output Considered	Independent Check	Action & Effect
Generative AI Assistant	2026-09-03	Initial report outline	Suggested treating $4.0B ARR as recognized revenue.	Checked Item 7 MD&A; verified ARR is unearned non-GAAP.	**MODIFIED**. Excluded ARR; retained recognized $707M.
Antigravity Coding Assistant	2026-09-24	Pro-forma BS logic	Proposed standard working capital formula with inventory.	Traced Form 10-K balance sheet; confirmed $0 inventory.	**REJECTED**. Replaced with Customer Prepayments modeling.
Antigravity Coding Assistant	2026-09-29	Sensitivity loop	Proposed parameter sweep on unarticulated FCFE schedule.	Checked course requirements for 3-statement articulation.	**MODIFIED**. Enforced 3-statement rebalancing (Gap=0.0000).
Antigravity Coding Assistant	2026-10-08	Audit & manifest	Diagnosed manifest parser error caused by unquoted commas.	Tested terminal execution with pandas and standard csv.	**ACCEPTED**. Produced valid quoted 12-row manifest.

6. Limitations, Monitoring, and Kill Rules

- **Kill Conditions:** Reject valuation immediately if balance sheet fails articulation (Gap > $0.01M), if cash buffer drops below $500M, or if terminal FCF is negative.
- **Observable Monitoring:** Track verified customer acceptance of Horizons 2–4 and positive operating cash flow before customer advances.